Native GLV-chain benchmark: Bandersnatch, Tom-384 and CP6-782 G1

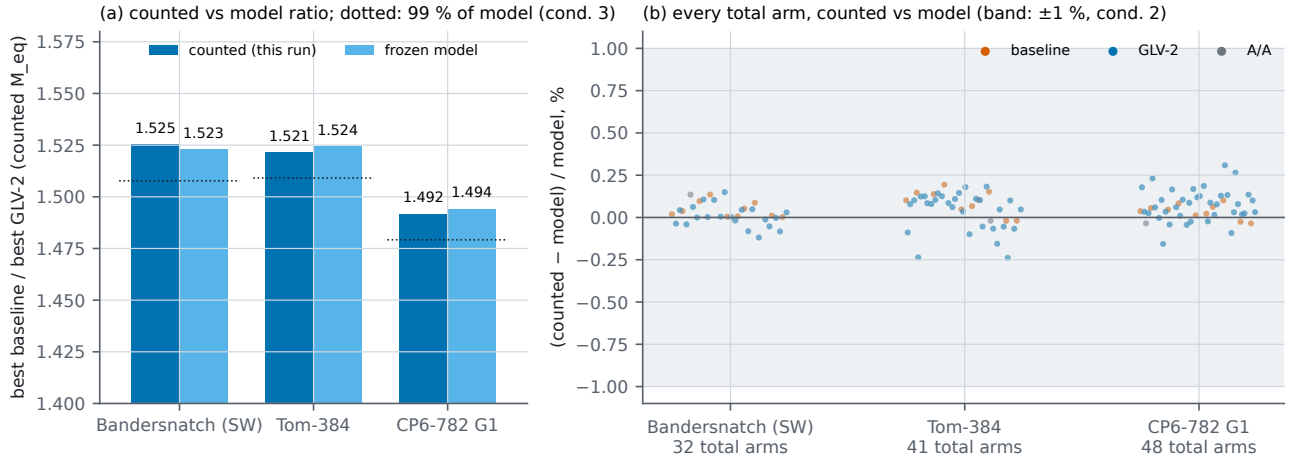
Results report · 2026-10-16 · `research/endo_native_bench_20261016/` · harness `85793758e`, protocol and runs `33e2dd10b` · PR #2006

Summary. GLV-2 through a cheap isogeny-chain endomorphism, measured natively with the curve-generic harness `endo-chain-bench` on the three curves the cross-curve sweep named next (`research/endosweep_curves_20261006`). Counted field operations ($M_{eq} = M + S$) are the primary result, wall time the secondary one. All five preregistered conditions of `PROTOCOL.md` are met on all three curves. Counted ratio best baseline / best GLV-2: **1.525** (Bandersnatch, short Weierstrass model), **1.521** (Tom-384), **1.492** (CP6-782 G1), against the frozen model's 1.523 / 1.524 / 1.494. Declared GLV arm against the fastest baseline by median: 1.504 / 1.504 / 1.570 (minimum ratios 1.528 / 1.530 / 1.520), A/A median ratios 0.997 / 1.001 / 1.038. Class **engineering** ($k \cdot P$ on the same curve, faster); variable-time harness; no security claim. Every number below is copied from the committed records and `README.md`; the figures are drawn only from the committed JSON records by `figures/plot.py`. The run report is `reports/run-20261016T1759Z/report.pdf`.

1. Preregistered conditions

condition	Bandersnatch	Tom-384	CP6-782 G1
1. correctness, both runs	every arm = textbook `BigUint` $k \cdot P$ on 256/256 pairs; $\varphi(P) = \lambda P$	yes	yes
2. $\varphi(P)$ counts exact; totals within 1 %	4/4 exact (declared 7M + 2S exception); worst +0.15 %	44/44; worst −0.24 %	114/114; worst +0.31 %
3. counted ratio ≥ 99 % of model	1.525 ≥ 1.508	1.521 ≥ 1.509	1.492 ≥ 1.479
4. median > 1.40, timed/counted within ±5 %	1.504; −1.3 %	1.504; −1.2 %	1.570; +4.7 % (inside 2× its 3.8 % A/A floor)
5. lowest $\varphi(P)$ median = optimised arm at least count	`0+1w/2`, 226 ns	`4+1w/7.5.5`, 7 652 ns	`4+1w/7.5.3`, 28 440 ns

2. Counted against the model

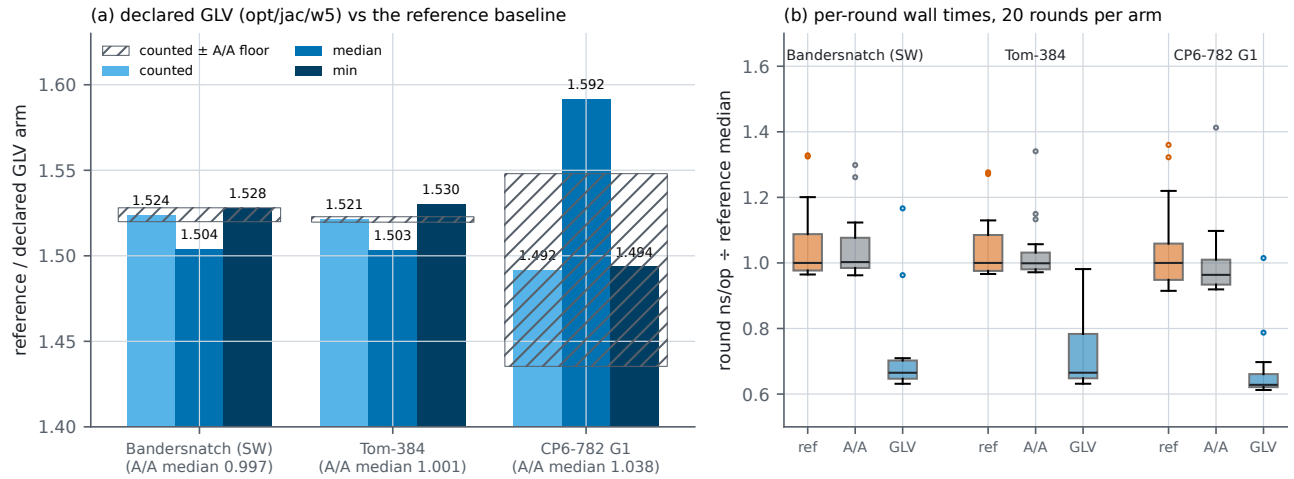


Counting runs (bandersnatch.counts.json, tom384.counts.json, cp6_782.counts.json), opcount build at commit 33e2dd10b; counted M_eq = M + S, mean over 256 pairs; model averages 400 other scalars

Figure 1: (a) Counted and model ratio, best baseline / best GLV-2, per curve, with the condition-3 threshold (99 % of model). (b) Deviation of every total arm's counted mean M_{eq} from the frozen model (32, 41 and 48 arms). Source: `<curve>.counts.json`, 256 pairs, seed 20261016.

The frozen model predicted every total arm from the same recoding and lattice bases, averaged over 400 other scalars; the counting runs land within 0.31 % on every arm. The model's counted ratios (1.523 / 1.524 / 1.494) and the counted ones (1.525 / 1.521 / 1.492) agree to within 0.2 %. The CryptoPro-B sweep measured 1.408 counted; GLV saves more on the general-a curves, whose doubling costs 2M + 8S, as the model predicts.

3. Timed, with the A/A floor

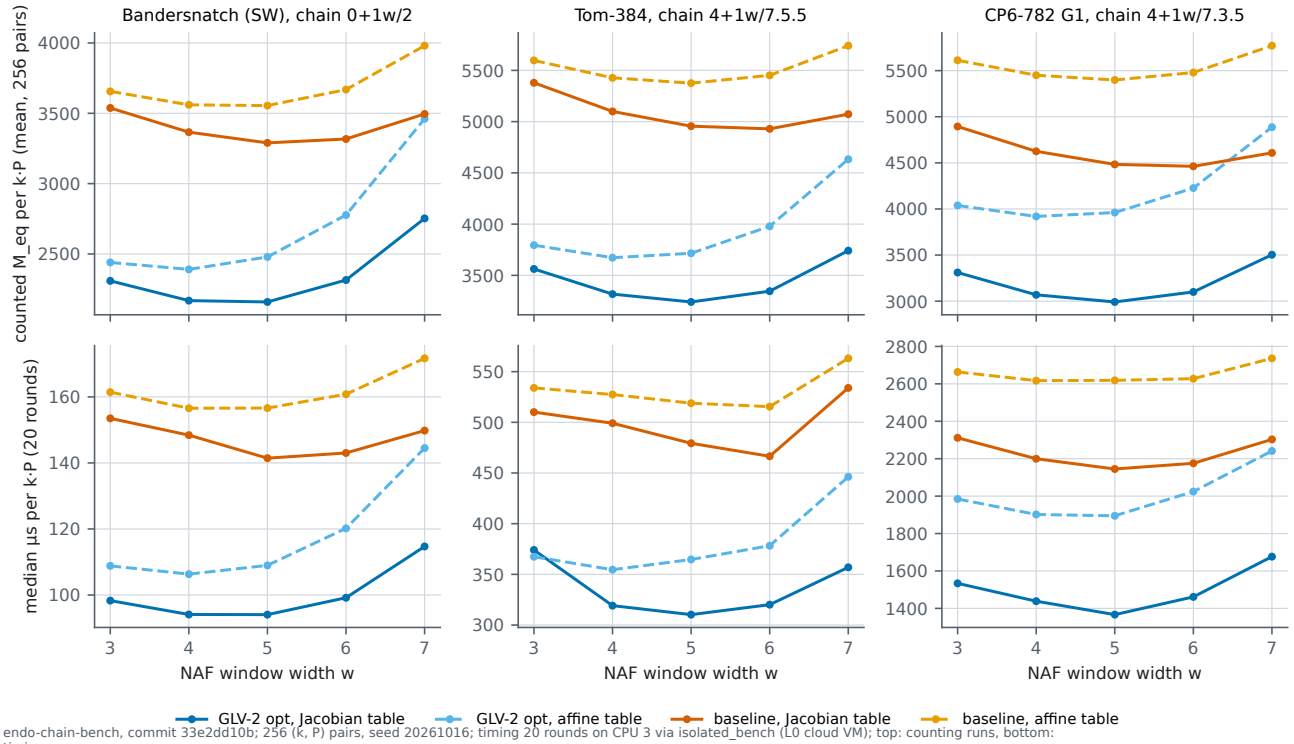


endo-chain-bench, commit 33e2dd10b; 256 (k, P) pairs, seed 20261016; timing 20 rounds on CPU 3 via isolated_bench (L0 cloud VM); ratios = recorded ratio_reference_over_arm against the reference baseline (Bandersnatch jacobian/w5, Tom-384 and CP6-782 jacobian/w6); CP6-782's fastest baseline by median is jacobian/w5 (README condition 4: 1.570)

Figure 2: (a) Declared GLV arm (`optimised/jacobian/w5` on the grid chain) against the reference baseline: counted, median and minimum ratio, hatched band = counted ratio ± the A/A floor $|1 - \mathrm{A/A\ median\ ratio}|$. (b) Per-round ns/op of the reference, the A/A arm and the declared GLV arm, normalised by the reference median (20 rounds each). Source: `<curve>.timing.json`.

The plotted ratios are against the reference baseline as recorded. On CP6-782 the fastest baseline by median is `jacobian/w5` (2 145 µs), not the reference `jacobian/w6` (2 175 µs); against it the declared arm's median ratio is 1.570 (condition 4), against the reference 1.592. CP6-782's timing run was the noisiest: A/A median ratio 1.038, per-round spread 49 % on the reference and 54 % on the A/A arm, 11.1 CPU-s of other activity in the 570 s run (admitted, `contended: false`, budget 0.5 CPU). Its wall time is read as consistent with its counts, not as a larger gain.

4. Width and table grid



endo-chain-bench, commit 33e2dd10b; 256 (k, P) pairs, seed 20261016; timing 20 rounds on CPU 3 via isolated_bench (L0 cloud VM); top: counting runs, bottom: timing runs

Figure 3: Counted M_{eq} (top) and median µs (bottom) per $k \cdot P$ against NAF width $w = 3..7$, for the declared chain (optimised evaluator) and the baseline, Jacobian and affine tables. Source: counting and timing JSON.

Jacobian tables beat affine tables for both arms at every width ≥ 4 on all three curves; the reason is the Fermat inversion (419, 684 and 1 173 M_{eq}). GLV's optimum is at $w = 5$ everywhere; the baseline's at 5 (Bandersnatch) and 6 (Tom-384, CP6-782) by count.

5. $\varphi(P)$ stage diagnostics

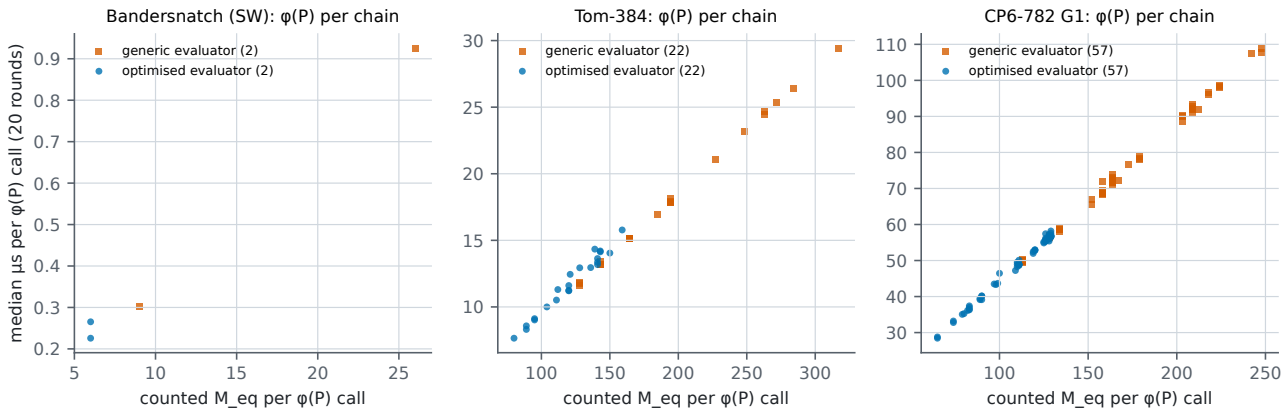


Figure 4: Every $\varphi(P)$ arm: counted M_{eq} per call against median µs per call, generic and optimised evaluators (2 + 2, 22 + 22, 57 + 57 chains). Stage diagnostics, never a speed of $k \cdot P$.

Every $\varphi(P)$ arm's counted M and S equals the chain set's model, 162 of 162, apart from the declared generic `0+1w/2` exception (7M + 2S counted, the model's 8M + 2S does not correspond to a correct evaluator). Spearman rank correlation of count and median: 0.949 (4 arms), 0.984 (44), 0.994 (114), as stated in `README.md`. The chain is 0.2–0.3 %, 2.5 % and 2.2 % of the declared GLV arm by count.

6. Scope and limits

- Bandersnatch is measured on its short Weierstrass model only; its deployed twisted-Edwards arithmetic is not measured.
- The baseline is a plain width-w NAF with the same Jacobian arithmetic, not a production library; the table conclusion depends on the Fermat inversion.
- Wall-time ratios hold within one curve only; no claim about cycles per field operation. Host: Intel Xeon @ 2.10 GHz cloud VM, 4 logical CPUs, timing pinned to CPU 3 through `isolated_bench` (L0, not core-isolated hardware).
- Chains outside the frozen chain sets are not measured. Positive control: the generic harness reproduces the CryptoPro-B sweep's committed counts on all 97 arms with 0 differences (`positive_control/`).

Records. `bandersnatch|tom384|cp6_782` × `.counts.json`, `.timing.json`, `.isolated_bench_record.jsonl`, with their `.stdout.md` tables; `frozen_inputs/` (chain sets, model, `SHA256SUMS`); `PROTOCOL.md`; `README.md`. Figures: `figures/plot.py` → `figures/fig_*.svg` / `.png`.